+ Lead Quality Analysis

DebtReductionInc Publisher x Debt-Settlement Advertiser, Apr to Sep 2009

A lead-generation case study: one publisher, one debt-settlement advertiser, about 3,021 leads sold between April and September 2009. This report answers three questions: is lead quality trending, what drives it, and how do we get to the advertiser's +20% quality target.

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More at: an interactive companion site covering all three questions (filterable trend view, segment explorer, opportunity sizing) is available at **https://markdechavez28.github.io/LeadQualityAnalysis/**. This PDF is the complete, standalone report; the site is a bonus exploration layer.

+ Executive Summary

Across the ~3,021 leads sold to the advertiser between April and September 2009, the blended Closed rate was **8.11%**, matching the advertiser's stated 8.0% baseline. Three findings stand out.

- **Lead quality is declining, and it's statistically significant.** The Closed rate fell from roughly 12-13% in early April to under 5% by late September (logistic regression p=0.007, Mann-Kendall p=0.013). Notably, the broader "Good lead" rate (Closed + EP-stage progress) shows *no* significant trend (p=0.81); the decline is concentrated in final closing, not in raw lead quality reaching the advertiser. That points toward a downstream sales/closing execution issue as much as a traffic-quality one.
- **Traffic source is the strongest quality driver we can act on.** Leads from Google's Display/Content Network ("Google-Display", identifiable by doubleclick.net referrals) close at just **4.1%**, less than half the rate of Google-Search (9.9%), AdKnowledge (12.3%), or Call Center (9.6%) leads, on a real volume of 639 leads (21% of the dataset). This survives a multivariate model controlling for widget, state, and debt level (p<0.001). Consumers with debt over $100k also close far less often (3.7% vs 7-14% elsewhere, p=0.043 controlling for other factors).
- **Redirecting Google-Display spend to Search-equivalent traffic, combined with excluding the >$100k-debt segment, gets to the 9.6% target**, projected blended rate of **9.75%**, holding total sold volume constant. This is the headline recommendation for the advertiser's +20%-quality-for-+20%-CPL offer.

See Methodology below for how "lead quality" is defined and handled, and the appendix for full statistical output.

+ Methodology Note

Lead-quality definition

The advertiser's stated baseline quality rate (8.0%) reconciles almost exactly with **Closed** leads (became a paying customer) as a share of all leads sold: 245 / 3,021 = 8.11%. We use **CallStatus == Closed** as the primary lead-quality metric throughout this report, since it is what the advertiser is actually offering to pay more for.

As a secondary, supporting cut, we also report a broader **Good** definition (Closed + EP Sent + EP Received + EP Confirmed, any forward progress toward becoming a customer) against a **Bad** definition (Unable to Contact + Contacted-Invalid Profile + Contacted-Doesn't Qualify). The remaining ~71% of leads have a blank CallStatus and are treated as **Unknown** (neither good nor bad), per the data dictionary; these are leads the advertiser has not yet resolved or reported back on. Where the Good/Bad cut would tell a materially different story than the Closed-only cut, we flag it explicitly (see Q1).

Data cleaning decisions

- **WidgetName size normalization:** ad sizes 300250 and 302252 are confirmed identical per the data dictionary; widget names were normalized to compare creative/design variants independent of size.
- **Partner field split:** the Partner column contains both "google" (lowercase) and "Google" (capitalized) as distinct values. Cross-referencing ReferralDomain shows this is not a casing typo: lowercase "google" is almost entirely www.google.com (organic/paid search), while "Google" is almost entirely googleads.g.doubleclick.net (Display/Content Network). We split these into **Google-Search** and **Google-Display** since they show a large, significant quality gap (see Q2) that would otherwise be hidden by merging them.
- **DebtLevel bucket harmonization:** the debt-amount bucketing scheme changed mid-dataset. April-May used a combined "7500-15000" bucket, while June onward split it into "7500-10000" and "10001-15000". We re-merged these into the coarser "7500-15000" bucket throughout so DebtLevel is comparable across the full date range.
- **PublisherZoneName / PublisherCampaignName / Call-Center Partner are perfectly collinear** in this dataset; the 271 Call Center leads are exactly the "Top Right-300x250" zone and exactly the "DebtReductionCallCenter" publisher campaign, no more and no less. Similarly, **AdvertiserCampaignName is perfectly collinear with the CreditSolutions widget variant**. We report each field's univariate relationship but collapse the redundant ones in the multivariate model (Q2) to avoid double-counting the same split three times.

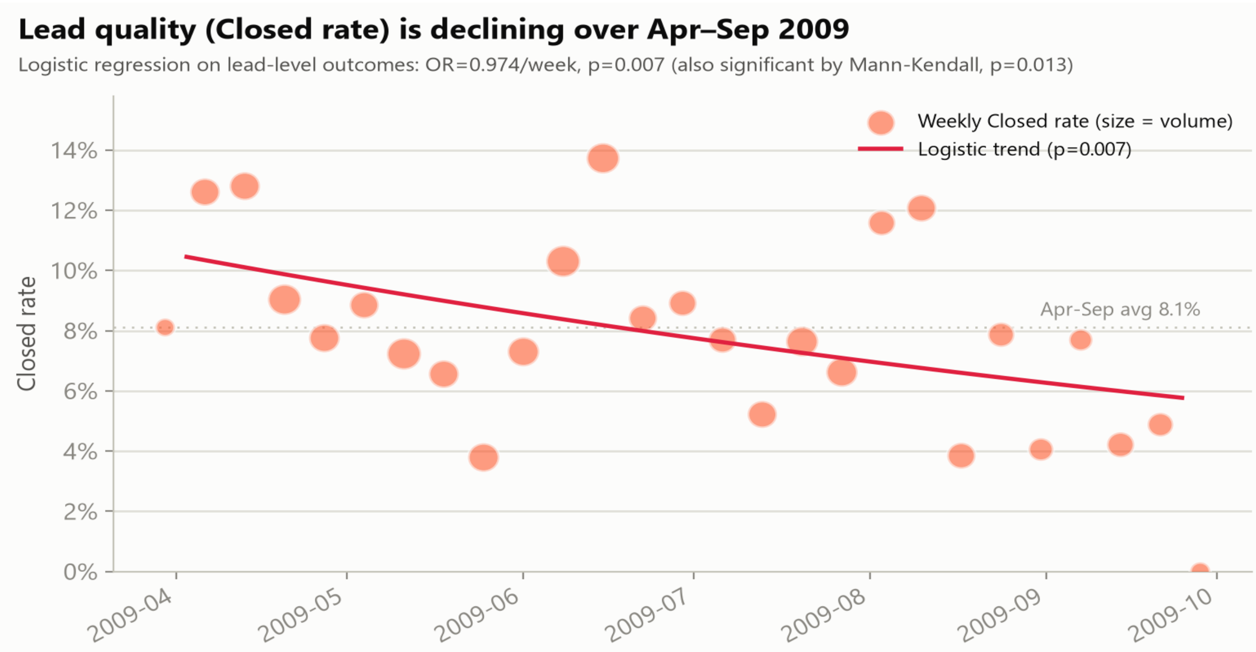
Data quality caveats

- **CallStatus is missing for 2,140 of 3,021 leads (70.8%).** These are treated as Unknown throughout. This is a large enough share that any rate we report should be read as "rate among resolved leads," not "rate among all leads sold"; we note this wherever it materially matters. Missingness is not evenly distributed (Call Center leads are resolved more often, 59% missing vs 72% for web-form leads), so naive comparisons that ignore Unknown status can be misleading across segments.

- **AddressScore and PhoneScore are only populated from 2009-06-26 onward** (1,393 of 3,021 leads, 46%). Any finding involving these fields is necessarily restricted to that later window and should not be assumed to hold for the full Apr-Sep period.
- **Small-n segments are flagged, not hidden.** Throughout Q2, segment levels with fewer than 30 leads are marked unreliable and excluded from chi-square tests, but shown in full tables in the appendix for transparency.
- **Multiple-comparisons risk:** Q2 scans many candidate segments (11 widget variants, 31 states, etc.) for the best/worst performer. With that many comparisons, some differences will look "significant" by chance. We call out this risk specifically for State, where the overall chi-square is not significant (p=0.54) even though individual states like OK and HI show high raw rates on modest n.
- We tested whether placeholder/refused emails (none@none.com, refused@refused.com, etc., 142 leads) predict worse quality; they do not (9.2% vs 8.1% Closed), so this was ruled out as a filtering lever.

+ Q1: Is Lead Quality Trending Over Time?

Yes, Closed rate is declining, and the decline is statistically significant. We tested this two ways: a lead-level logistic regression of Closed on week number (the primary test, since it uses every individual outcome rather than collapsing to weekly averages first), and a Mann-Kendall non-parametric trend test on the weekly aggregated rate series as a distribution-free cross-check.



Weekly Closed rate, Apr-Sep 2009. Point size reflects weekly lead volume. Red line is the fitted logistic trend.

Test	Statistic	Result	p-value
Logistic regression (Closed ~ week)	OR = 0.974 / week (95% CI 0.956–0.993)	Odds of closing fall ~2.6%/week; ~48% lower by week 25 vs week 0	0.0072
Mann-Kendall (weekly rate series)	S = –120, z = –2.48	Decreasing trend confirmed non-parametrically	0.0131
Secondary check: Good rate ~ week	OR = 1.002 / week	No significant trend in the broader Good definition	0.807

The nuance worth flagging: **the decline only shows up in the Closed rate, not in the broader Good rate** (Closed + EP stages). If the traffic mix or lead quality reaching the advertiser were simply getting worse, we'd expect both metrics to fall together. Instead, leads appear to be advancing through the EP pipeline at a roughly stable rate over time, but converting to Closed less often as the year progresses. That's more consistent with a downstream sales/closing execution change (call-center staffing, worksheet follow-through, advertiser capacity) than with declining lead quality at the point of sale. It's worth investigating operationally alongside the traffic-composition levers in Q3.

Monthly Closed rate: Apr 10.8% → May 6.4% → Jun 10.3% → Jul 6.2% → Aug 9.4% → Sep 4.4%. The month-to-month swings are large, but the week-level regression and Mann-Kendall test both confirm a genuine downward drift underneath that noise, not just random variation.

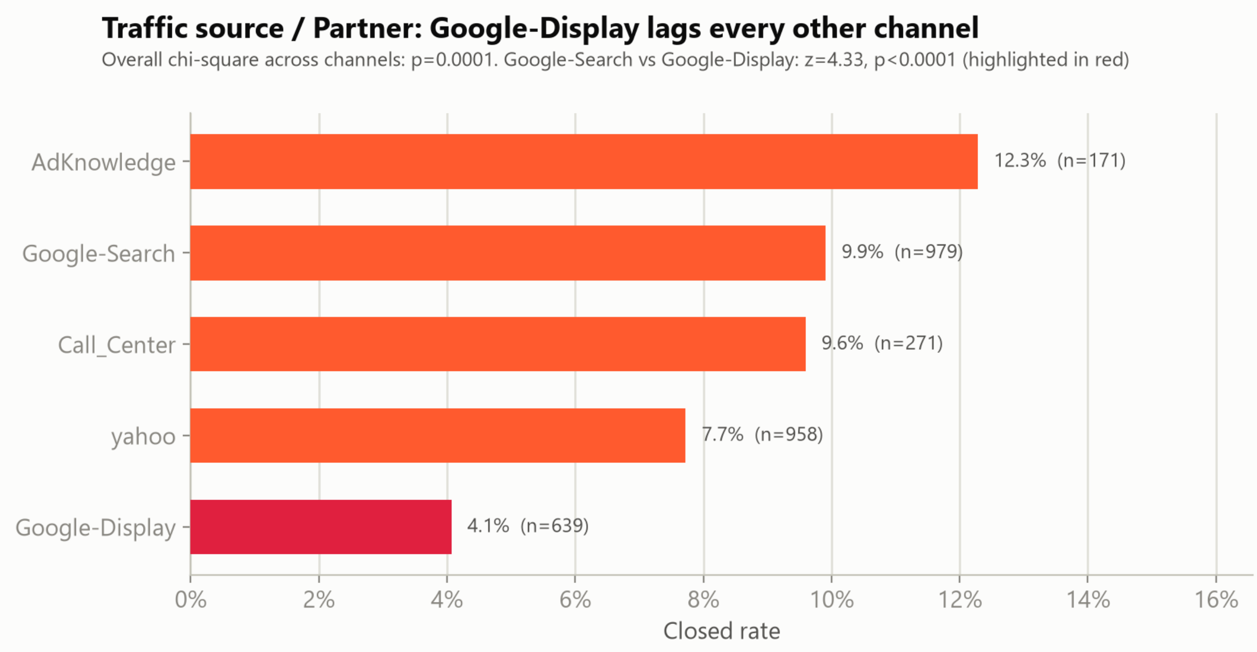
+ Q2: What Drives Lead Quality?

We tested each candidate segment two ways: a chi-square test of independence across all reliable (n≥30) levels, and, for the top drivers, a multivariate logistic regression that controls for the other segments simultaneously (since several are correlated, e.g. Call Center leads cluster on specific publisher zones by construction).

Ranked driver summary

Segment	Chi-square p-value	Survives multivariate control?	Verdict
Partner / traffic source	0.0001	Yes: Google-Display p<0.001	**Strong, actionable driver**
Debt level	0.091 (overall); >$100k tier significant	Yes: p=0.043	**Strong, actionable driver**
PhoneScore	0.052 (categorical); continuous term significant	Yes: p=0.039 (post-Jun subset)	Supporting signal, smaller effect
WidgetName / creative	0.418	No	No reliable effect detected
PublisherZone / Campaign (web vs call-center)	0.411	Collinear with Partner=Call_Center	No independent effect beyond Partner
AdvertiserCampaignName (branded vs generic)	0.653	Collinear with widget	No effect
State	0.538 (overall)	Borderline for 1-2 states only	Not reliable, multiple-comparison risk
AddressScore	0.322	Not tested (dropped for PhoneScore)	No clean signal

Traffic source / Partner



Closed rate by traffic source, n≥30 only. Advertise.com (n=3) excluded as unreliable.

Google-Display leads close at 4.1% vs 9.9% for Google-Search, a statistically significant gap (two-proportion z-test: $z=4.33$, $p<0.0001$) that holds up after controlling for widget, state, and debt level in the multivariate model (OR=0.31, 95% CI 0.17–0.58, $p<0.001$, vs AdKnowledge as reference). It also holds under the broader Good-rate definition (7.0% for Google-Display vs 13.8-18.7% for other channels), so this isn't an artifact of the Closed-only metric. Interestingly, Google-Display's *bad*-lead rate (15.0%) is not much worse than other channels (13-24%); its problem is specifically weak downstream conversion, consistent with lower purchase intent from display/content-network impressions versus active search intent, not a data-quality or fraud issue.

Debt level

Closed rate by consumer-reported debt level (buckets harmonized, see Methodology).

Consumers reporting over $100k in debt close at just 3.7%, the lowest of any debt tier, and the only tier with a below-baseline rate that's also statistically significant in the multivariate model (p=0.043) after controlling for widget, partner, and state. This tier also has the highest bad-lead rate (19.9%), so the pattern is consistent across both definitions: these are less qualified prospects, plausibly because a debt-settlement program isn't a realistic fit for extremely high balances, or because these leads skew toward unemployment/inability-to-pay (a disqualifying reason per the data dictionary).

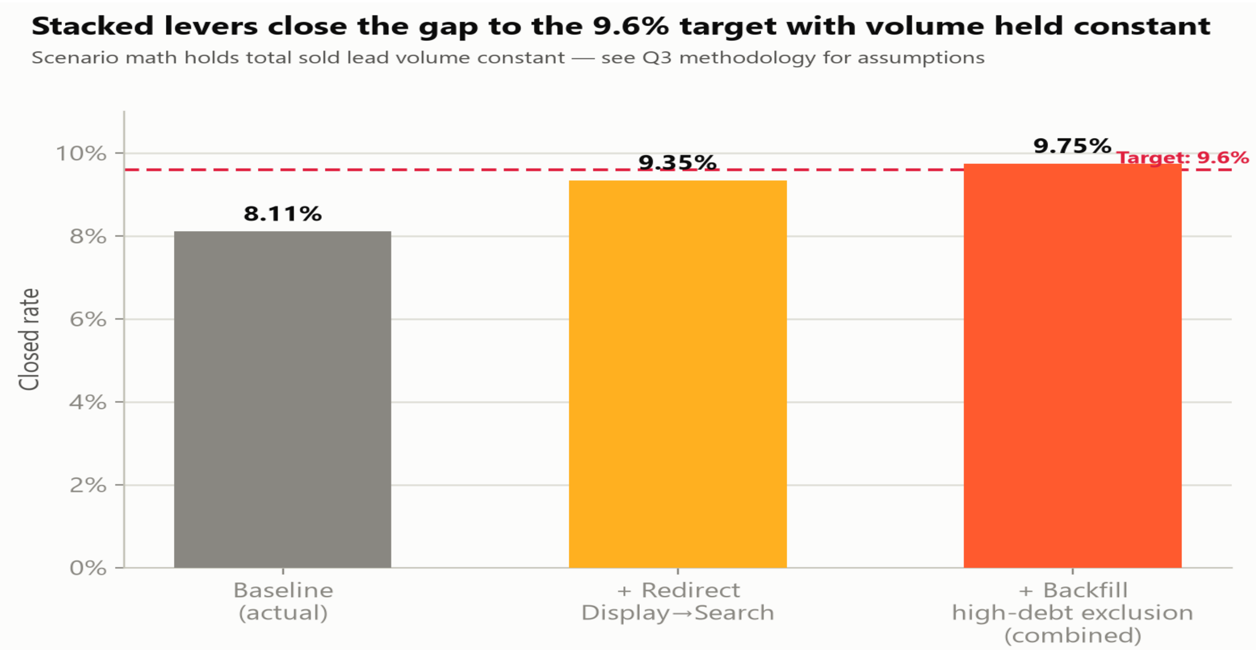
Other segments tested

WidgetName / creative design shows no statistically reliable effect (chi-square p=0.42) despite some spread in raw rates (1DC-BlueMeter 14.1%, n=92 vs 1DC-Head3 5.3%, n=75); with 11 variants compared, that spread is well within what multiple comparisons on modest sample sizes would produce by chance. **PublisherZoneName/PublisherCampaignName** (page placement, web-form vs call-center) shows no effect once you account for the fact it's identical to the Call_Center partner split already tested above. **AdvertiserCampaignName** (branded vs generic ad) shows no effect (p=0.65) and is collinear with the CreditSolutions widget variant. **State** is not significant overall (chi-square p=0.54 across 22 reliable states); a few states like Oklahoma (18.0%, n=50) and Hawaii (15.6%, n=32) look attractive individually, but with 22+ states tested this is exactly the kind of pattern that shows up by chance; we don't recommend acting on state alone. **PhoneScore** shows a modest, directionally consistent signal (OR=1.24 per point, p=0.039 in the multivariate model restricted to the post-June scored subsample), higher phone-match confidence associates with better quality, but the effect is smaller than traffic source or debt level. **AddressScore** shows no clean monotonic relationship (p=0.32) and is not a reliable lever.

+ Q3: Path to +20% Lead Quality (8.0% to 9.6%)

The advertiser's offer requires moving the blended Closed rate from 8.0% to 9.6%, a gap of 1.49 percentage points. The two significant, actionable drivers from Q2, traffic source and debt level, get there when stacked together, with total sold lead volume held constant.

BASELINE (ACTUAL)	PROJECTED (COMBINED LEVERS)
8.11%	**9.75%**



Scenario sizing: each bar assumes total sold volume is held at 3,021 leads; see notes below for exact assumptions.

Lever	Mechanism	Projected rate	Lift	% of gap closed
1. Redirect Google-Display budget to Search	639 Display leads re-sourced via Search at Search's observed 9.9% rate, same volume	9.35%	+1.24pp	83%
2b. Exclude >$100k-debt leads, backfill volume	191 high-debt leads replaced with volume at the 8.41% blended rate of all other tiers	8.41%	+0.30pp	20%
3. Combined (1 + 2b)	Both levers stacked, same total volume	**9.75%**	**+1.64pp**	**110%**

Recommended levers

- **Reallocate Google ad spend from Display/Content Network toward Search.** This is the single highest-leverage move available; it alone closes 83% of the gap. It requires no changes to the publisher's widgets, forms, or the advertiser relationship; it's a media-buying decision within the team's direct control. Practically: audit the current Google campaign structure for content-network/GDN placements (identifiable by doubleclick.net referral domains) and shift that budget into search-targeted campaigns, which already show 2.4x the close rate on comparable volume.

- **Add a debt-level qualification gate before selling >$100k-debt leads,** or route them to a different buyer/program better suited to very high balances. Combined with the traffic reallocation, this pushes blended quality past the 9.6% target with room to spare (9.75% projected, 110% of the required gap). Note this reduces the pool of leads eligible for this advertiser by about 6%; the backfill assumption in scenario 2b assumes that volume is replaced with additional traffic at the blended rate of the remaining pool, a reasonable assumption if publisher volume isn't already capacity-constrained.
- **Investigate the Q1 closing-execution decline as a separate, unsized opportunity.** Since the Good-rate trend is flat while Closed-rate is falling, there may be upside available purely from improving downstream follow-through (worksheet completion reminders, call-center capacity, faster confirmation turnaround) that wouldn't require any traffic changes at all. We can't size this precisely from this dataset since it's a process question, not a composition question, but it's worth an operational review in parallel with the two levers above.
- **PhoneScore-based pre-sale filtering is a smaller, supporting lever** worth piloting once the score is available on more of the volume: leads with PhoneScore 4-5 close at 7.6% vs 5.9% for PhoneScore 2-3 within the post-June scored subsample. The effect is real (p=0.039 in the multivariate model) but modest relative to the two levers above, and currently only covers 46% of volume.

A caution on precision: these scenarios reallocate historical averages forward: they assume Google-Search and the non-high-debt pool would maintain their observed rates at higher volume, which may not hold exactly if diminishing returns or audience saturation kick in at scale. We'd recommend testing the traffic reallocation on a partial-budget basis first and re-measuring before committing fully.

+ Conclusion

Putting the three questions together: lead quality is worth watching closely right now. It's trending down over the six months in this dataset, but the cause looks fixable rather than structural. Two straightforward changes to where traffic comes from and which leads get sold can plausibly clear the advertiser's quality target on their own, with no changes needed to the publisher's widgets, forms, or the advertiser relationship.

In plain terms: buy less Google Display traffic and more Google Search traffic, stop selling leads from consumers carrying over $100k in debt, and keep an eye on whether the sales team's closing rate recovers on its own over the next few months. Doing the first two already clears the advertiser's 9.6% bar on paper; doing all three leaves room to beat it.

What this means in practice

- **This quarter:** shift a meaningful share of Google ad spend from Display/Content Network to Search, and add a debt-level check before selling leads over $100k. Both are media-buying and qualification decisions within the team's direct control.
- **Ongoing:** track Closed rate and Good rate side by side every month. If Good rate stays flat while Closed rate keeps falling, that confirms the issue is downstream, in closing execution, not in the leads themselves, and the fix is operational rather than traffic-related.
- **As data accumulates:** PhoneScore is a promising secondary filter, but it is only populated for about half of current volume. Once it covers more of the traffic, revisit it as a pre-sale quality gate.

For a closer look at any of these numbers, including a segment-by-segment breakdown you can filter yourself, see the interactive companion site at **https://markdechavez28.github.io/LeadQualityAnalysis/**.

+ Appendix

A1. Overall quality-definition breakdown

Category	n	% of total
Closed	245	8.11%
Good (Closed + EP Sent/Received/Confirmed)	393	13.01%
Bad (Unable to Contact / Invalid Profile / Doesn't Qualify)	488	16.15%
Unknown (blank CallStatus)	2,140	70.84%
Total leads sold	3,021	100%

A2. Multivariate logistic regression: key coefficients

Model: Closed ~ Widget variant + Partner (traffic source) + State (top states + Other) + Debt level. n=3,021. Reference categories: widget=1DC (base), partner=AdKnowledge, state=AL, debt=15001-20000.

Predictor	Odds ratio	95% CI	p-value
Google-Display (vs AdKnowledge)	0.31	0.17 – 0.58	<0.001
yahoo (vs AdKnowledge)	0.60	0.35 – 1.01	0.053
Google-Search (vs AdKnowledge)	0.83	0.50 – 1.37	0.413
Call Center (vs AdKnowledge)	0.68	0.35 – 1.35	0.275
Debt >$100k (vs 15001-20000)	0.42	0.18 – 0.97	0.043
Debt 70001-90000 (vs 15001-20000)	1.72	0.93 – 3.18	0.085
New York state (vs top-state baseline)	0.45	0.20 – 1.00	0.049
Widget variants (all)	0.59 - 1.63	n/a	all p > 0.16 (n.s.)

Robustness check adding PhoneScore (restricted to the 1,393 leads with a score, i.e. created 2009-06-26 or later): PhoneScore OR=1.24 per point, p=0.039, directionally consistent with the univariate pattern, controlling for widget/partner/state/debt.

A3. Full segment tables (Closed rate)

Widget variant	n	Closed rate
1DC-BlueMeter	92	14.1%
1DC-Head2	89	12.4%
1DC (base)	620	9.2%
2DC-CreditSolutions	75	8.0%

Widget variant	n	Closed rate
1DC-CreditSolutions	1,131	7.8%
1DC-yellowarrow-dark	135	7.4%
1DC-white	436	7.3%
2DC-BlueMeter	87	6.9%
1DC-yellowarrow-blue	232	6.5%
1DC-yellowarrow	49	6.1%
1DC-Head3	75	5.3%

Debt level	n	Closed rate	Good rate	Bad rate
7500-15000	1,004	7.5%	11.6%	18.3%
15001-20000	408	8.6%	14.0%	12.0%
20001-30000	456	8.8%	15.8%	15.4%
30001-50000	496	7.9%	12.3%	16.9%
50001-70000	245	9.0%	15.5%	13.9%
70001-90000	131	13.7%	19.1%	12.2%
90000-100000	90	10.0%	12.2%	14.4%
More than 100000	191	3.7%	6.8%	19.9%

Full row-level analysis code (Python/pandas/statsmodels/scipy), all intermediate output tables, and chart-generation scripts are included alongside this report as supporting/appendix material.